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SPECIFICATION FOR DC BRUSHLESS MOTOR

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REV					
	APPROVED	Y.SEKII	2015-01-26	MODEL	A2240N02
	DESIGNED	A.ITO	2015-01-26		
APPROVED		Y.SEKII	2013-11-13	DRAWING No.	1NSPC136001D
CHECKED		T.IGUCHI	2013-11-13		
DESIGNED		A.ITO	2013-09-09	DC BRUSHLESS MOTOR	Sheet 1 of 9
DRAWN		A.ITO	2013-09-09		

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SPECIFICATION FOR DC BRUSHLESS MOTOR

1. Purpose

This document defines th e motor requirements for spinning color for filter wheel in DLP projector.

2. Specification

2-1. General Specification

No.	Item	Specification	Notes
1	Supply voltage	DC 36 [V] +/-5%	
2	Type	8 poles, 3 phases, brush less, outer rotor sensor less DC motor.	
3	Speed of rotation	10800 [min^{-1}]	
4	Direction of rotation	CCW *When starting the motor, it is normal to have CW rotation, then switch to CCW.	When viewed from hub side. No reverse rotation due to damage on bearing.
5	Load (Color wheel)	Unbalance : G6.3 MAX Mass : 100 [g] MAX Inertia : 1000 [$g\text{-}cm^2$]	With Motor / At 10800[min^{-1}] Include color wheel hub and balancing weight. The gravity center of color wheel assembly should be positioned between 7.5mm and 10.5mm above the hub flange surface as 21.75mm.
6	Set position	No limitation	
7	Mass	85 [g] MAX	Motor only
8	Motor Inertia (Rotor assembly)	Mass : 36.4 [g] Inertia : 68.3 [$g\text{-}cm^2$]	As nominal value
9	Starting Current	1.2 [A] +/-8% (In the setting of the circuit board of an actual device) Temperature: 5-85 [deg-C]	Nidec recommends the customer to set the current limit at the motor coil line to 1.1[A] or more for the motor start up after the low temperature and the long preservation. Do not supply the current of 1.0A or more for 1 minute or more.

2-2. Electrical parameters (At 20 deg.C, normal humidity)

No.	Item	Specification	Notes
1	No Load Current	230 [mA] MAX	Speed: 10800[min^{-1}], Coil current only, After the motor started, measured one minute later. At 20deg.C, normal humidity
2	K_T(Torque constant)	22 [mNm/A] +/-10%	Rms.
3	Terminal Resistance	2.25 [ohm] +/-15%	Phase to phase At 20deg.C, normal humidity
4	Terminal Resistance	1.2 [ohm] +/-15%	Phase to common At 20deg.C, normal humidity
5	Inductance	1.5 [mH] MAX	1[kHz], Phase to phase, Avg.
6	Insulation resistance	10 [MΩ] MIN (At DC 250 [V])	Check between FPC and bracket.

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7	Hi-pot	1 [s] MIN (At AC 300 [V], 1[mA])	Check between FPC and b racket.

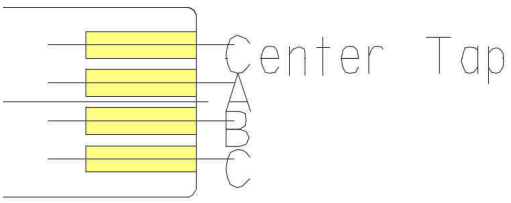
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2-3. Mechanical parameters (At 20 deg.C, normal humidity)

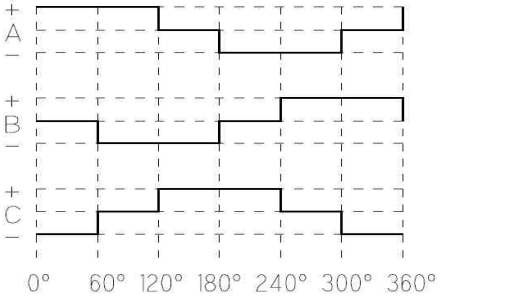
No.	Item	Specification	Notes
1	Noise 	55 [dB(A)] MAX (Initial Value)	Speed: 10800 [min-1], No load, Shaft vertical, on sponge, Microphone at 4 [cm] aligned on the motor axis. Frequency range: 0-20 [kHz]. Motor should be set on 10mm thick Al base. Check no vibration of chattering between FPC and the sponge. The background noise at measuring environmental is to be margin 10[dB(A)] min from motor noise.
2	Unbalance	80.0 [mg-cm] MAX	Motor only, At one side At 10800 [min-1]
3	Axial Play	0.05 [mm] MAX	
4	Vibration (G-Value) 	130 [mm/s2]	Motor only, At 10800 [min-1] *Vibration is measured at 180Hz related to unbalance. The motor is put up to a special fixture with the vibration sensor. The value of vibration is the addition of primary, secondary and third-order of rotation speed of FFT wave-form.

* Characteristics of motor are guaranteed at 10800[min-1]

2-4.Electrical interface



2-5.Timing chart



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3. Environmental Conditions

3-1. Check conditions

The check conditions should be ke pt 24 hours MIN at normal temperature and humidity.

3-2. Test conditions

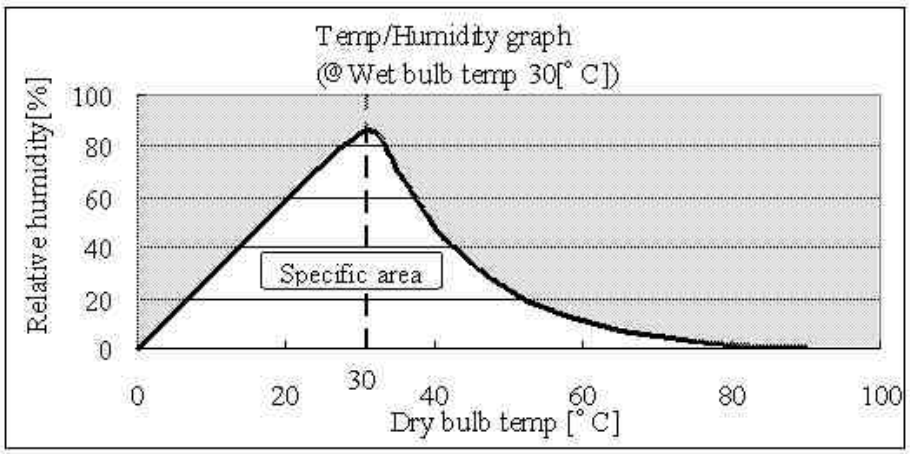
No.	Item	Specification	Test condition	Judgment
1	Operating	Dry bulb temp.: 5~85 [deg.C] Wet bulb temp.:30deg.C MAX Relative humidity: 5~90[%] (Humidity condition is to be refer to 3-2. No condense.) Altitude: -300~3,000[m] Maximum allowed temperature gradient is 20[deg.C/Hr]. Maximum allowed humidity gradient is 20[%/Hr].	10800 [min^{-1}], No load	It should be specified by item 2parameters expect noise. Noise should be 5[dB(A)] MAX from initial value.
2	Non-operating	Dry bulb temp.: -40~85 [deg.C] Wet bulb temp.:30 [deg.C] MAX Relative humidity: 5~90[%] (Humidity condition is to be refer to 3-2. No condense.) Altitude: -300~12,000 [m] Maximum allowed temperature grad ient is 30[deg.C/Hr]. Maximum allowed humidity gradient is 20[%/Hr].		It should be specified by item 2 parameters expect noise. Noise should be 5 [dB(A)] MAX from initial value.
3	Vibration	10-500[Hz], 5[G]	Apply Axis : X, Y, Z Speed : 1[oct/min] Apply 4[min] x 4 cycle per orientation	It should be sp ecified by item 2 parameters expect noise. Noise should be 5 [dB(A)] MAX from initial value.
4	Shock	Half-sine, 120[G's] / 2[msec]	Apply Axis : X, Y, Z Number : 5 drops per orientation	It should be specified by item 2 parameters expect noise. Noise should be 5 [dB(A)] MAX from initial value.

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3-2. Humidity condition

The worst environment is under high temperature/humidity, i.e. at 30 [deg.C] and 100[%] humidity. Wet bulb temperature is to be set as under 30 [deg.C].

Therefore, relative humidity changes by dry bulb temperature as following shows, and environment al condition should be within the specific area.



3-3. Fumigation

Exposure of this product's bearing to fumigate gas may cause corrosion, which may affect the motor's characteristics and reliability. Therefore, the motor's characteristics cannot be guaranteed if any wooden packing materials are fumigated together with packing boxes containing the product.

The product equipped with the motor must confirm whether the transportation route beforehand does the fumigation processing.

4. Life

4-1. Check Condition

The check conditions should be kept 24 hours min at normal temperature and humidity.

4-2. Spec and Test conditions

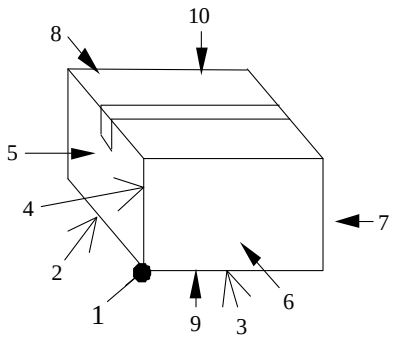
No.	Item	Specification	Notes	Judgment
1	B10 (Failure rate 10[%])	30,000[Hr] [@85deg.-C] 00D*1	Temperature: 5-85 [deg-C] Humidity :normal (No condensation) 10800[min^{-1}] With rated load (100[g])	*When load current changed 30 [%] MIN from initial value. *Noise should be 5[dB(A)] MAX from initial value. *Motor vibration with no load condition should be 15 [mm/s^2] MAX from initial value. However, in case the noise level and load current is satisfied spec, the time that exceed the spec would be the motor life.

*About the failure in time of guarantee, discuss sincerity each other and make clear share of responsibility.

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5. Packing

No.	Item	Specification	Notes	Judgment
1	Packing	Must meet level II in JIS-Z0200.	*Drop height: Drop from 60[cm] +/-2% *Drop procedure: Drop the package to the concrete ground one time on each side as following order. 1 corner/ 3 edges/ 6 sides Corner and edge should be dropped as the gravity line of the package pass through the corner or edge. The side should be dropped as the all side hit the ground. The order of the dropping is as the left picture. ex)1-(2-4-3)-(10-8-9-6-7-5)	*It should be specified by Item 2parameters except noise *Noise should be 5[dB(A)] MAX from initial value.



6. Full view drawing

6-1. Drawing

Refer to outer appearance drawings : 6023077300B

6-2. Serial number

- Barcode : micro QR code ver.M2
- Code detail

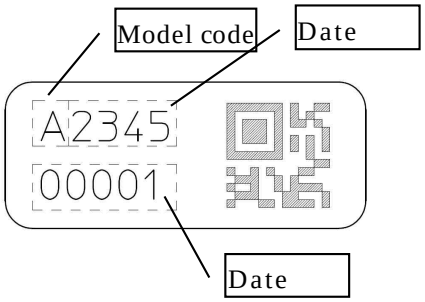
Model code : B : 2240N02

Date code : 2 : Year 3 = 2013, 4=2014, --- , 9=2019, 0=2020, 1=2021

34 : Week 1 ~ 52 (A week containing Jan -4 is defined as the 1 st week)

5 : Day : 1=Mon, 2=Tue, 3=Wed, 4=Thu, 5=Fri, 6=Sat, 7=Sun

Serial number : 00001 ~ 99999



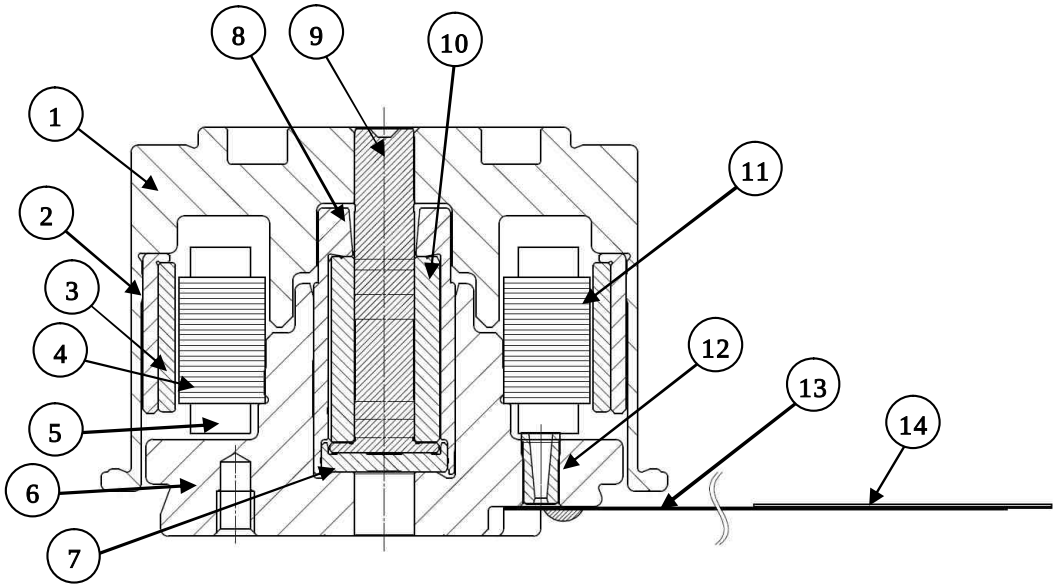
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SPECIFICATION FOR DC BRUSHLESS MOTOR

7. Parts list

No.	Parts	Material or Type	UL		Manufacture
			Grade	File No.	
1	Hub	A6061	-	-	
2	Back Iron	SUS430, QS430FT			
3	Magnet	Ne-Fe-B Isotropy	-	-	
4	Stator Lamination	35A250	-	-	
5	Copper wire	2-SF.BY	F TYPE	E339330	-FEMM
		2-FLW-N-OL	F TYPE	E82222	-SUMITOMO WINTEC
		PN180	H TYPE	E258243	-ELEKTRISOLA
		2-SF.BLOCKY3H	F TYPE	E339330	-FEMM
6	Bracket	A6061	-	-	
7	Thrust Bush	C6801, C6802	-	-	
8	Housing	C6801, C6802	-	-	
9	Shaft	SUS420	-	-	
10	Sleeve	Sintered Metal	-	-	
11	Core Coating	EX-1101	B TYPE	E98667	-Sumitomo Bakelite
		EL-1000	B TYPE	E78776	-TOA GOSEI
12	Insulation Bush	201NF (XFR4840)	94V-0	E213445	-WINTECH POLYMER LTD
		CM1014-V0	94V-0	E41797	-TORAY INDUSTRIES INC
13	FPC	Polyimide	94V-0	E252688	-Flex Plus
14	Barcode Label	PET			



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SPECIFICATION FOR DC BRUSHLESS MOTOR

8. Change history

Date	Rev.	No.	Before	Design	Approved
2014/3/24	00A	1	Addition of comment	A.ITO	Y.SEKII
		2	T.B.D.		
		3	6023077300A		
2014/3/25	00B	1	5[mm/s ^{2}]	A.ITO	Y.SEKII
2014/12/19	00C	1	10,000[Hr]	A.ITO	Y.SEKII
2015/1/26	00D	1	20,000[Hr]	A.ITO	Y.SEKII